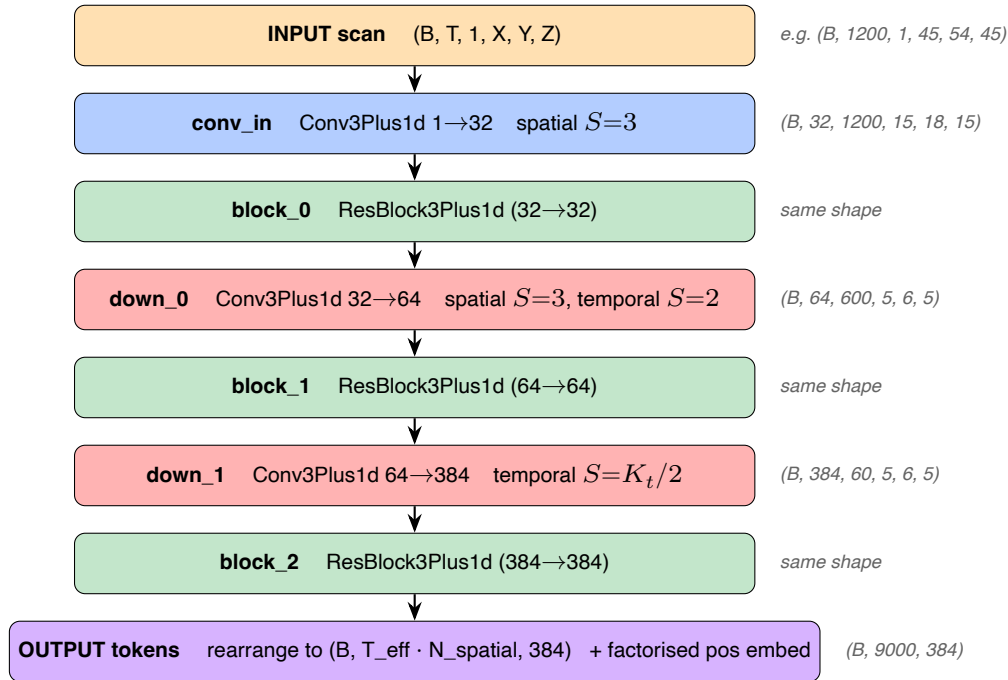


Patchify architecture — PatchEmbed3DPlus1D

2026-06-14

Hierarchical 3D+1D patchify encoder, inspired by MovieGen TAE (github.com/MathieuTuli/MovieGen, `tae.py:740`). Takes a 4D fMRI scan $(B, T, 1, X, Y, Z)$ and outputs a flat sequence of $T_{\text{eff}} \times N_{\text{spatial}}$ tokens of dim 384, consumed by the ViT.

Building block = Conv3Plus1d (separable conv): Conv3d spatial (per frame) + Conv1d temporal (per voxel-position). PyTorch lacks a native Conv4d, so we replace it with this separable variant — same trick as MovieGen's Conv2Plus1d, $\sim K \times$ fewer parameters than a full Conv4d.



Total strides — spatial $3 \cdot 3 \cdot 1 = 9$ (= patch_size), temporal $1 \cdot 2 \cdot K_t/2 = K_t$ (= temporal_kernel). Token grid is then $(T_{\text{eff}}, N_x, N_y, N_z)$ flattened.

Factorised positional embedding (before ViT) — three small tables broadcast and summed: $\text{pos_temporal}(T_{\text{eff}}) + \text{pos_spatial}(N_{\text{spatial}}) + \text{pos_cls}(1)$. For HCP setup: $O(60 + 150) \cdot 384 = 80, \text{k params}$ vs $O(60 \cdot 150) \cdot 384 = 3.46, \text{M}$ for a flat 2D pos table ($\sim 43 \times$ fewer).

Two configurations —

Setup	T	K_t	T_{eff}	N_{spatial}	Total tokens
HCP-YA	1200	20	60	$5 \cdot 6 \cdot 5 = 150$	9000
Mixed-T140	140	14	10	150	1500

File: `dinov2/layers/patch_embed_3d_plus_1d.py` (~230 lines, 4 classes: Conv3Plus1d, _ResBlock3Plus1d, PositionEmbedding3D, PatchEmbed3DPlus1D).